

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2019-2020 (Even Semester)
Innovative Practices Description for Unit-I

Degree, Semester & Branch: VI Semester B.E. ECE B

Course Code & Title: EC8691 Microprocessors and Microcontrollers

Name of the Faculty member: Mr.B.Kannan

Name of the Topic	Name of the Innovative Practice	Date & Duration
8086 Microprocessor Architecture & Instruction Set of 8086	Zero Minute Speech	21.01.2020 (10 Minutes)
Assembler Directives	One Minute Paper	04.01.2020 (10 Minutes)
Interrupts and Interrupt Service Routines	Animation Video	11.01.2020 (10 Minutes)

Description:

(i) Zero Minute Speech:

- The activity Zero Minute Speech is an impromptu speech where the speaker is supposed to express the idea(s) on the given topic, within a minute.
- The important topics were allotted to set of students on spot and they will give a speech on the topic with doubt clarification among classmates.

(ii) One Minute Paper:

- A “one-minute paper” may be defined as a very short, in-class writing activity (taking one-minute or less to complete) in response to an instructor-posed question.
- It prompts students to reflect on the day’s lesson and provides the instructor with useful feedback.

(iii) Animation video

- The use of audio-visual aids helps in maintaining discipline in the class since all the students attention is focused on learning. This interactive session also develops critical thinking and reasoning that are important components of the teaching-learning process.
- Traditional verbal instructions can be boring and painful for students. However, the use of audio-visual provides intrinsic motivation to students by improving their inquisitiveness and stimulating their interests in the subjects.

Goals (Learning Outcomes):

- The students will be able to describe the architecture of microprocessor 8086.
- The students will be able to gain knowledge on assembler directives.
- The students will be able to explain about interrupts and interrupt service routines.

Use of appropriate method:

Justification for choosing the following Activities:

- **Zero Minute Speech:** Leads to express the basics of the particular topics by different students.
- **Minute Paper:** The great advantage of minute papers is that they provide manageable amounts of timely and useful feedback for a minimal investment of time and energy.
- **Audio & Video Tools:** Audio-visual provides opportunities for effective communication between teachers and students in learning.

Effective Presentation: (Implementation (Plan & Execution) with Proof):

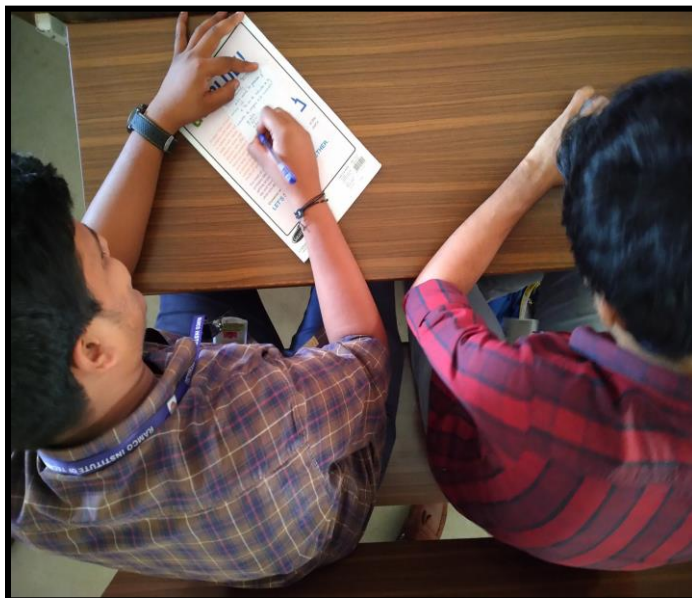
- **Zero Minute Speech:** It does not need a big preparation so I can involve the slow-learners also.
- **Minute Paper:** Students used small paper to write a few points about assembler directives.
- **Animation:** Students can easily understand the interrupt request from I/O devices and 8086 microprocessor response to that interrupt through animation video and also give some input to recall the information from their conventional learning.

**Zero Minute Speech for 8086 Microprocessor Architecture & Instruction Set
on 21.01.2020 (10 Minutes)**



One Minute Paper for Assembler Directives on 04.01.2020 (10 Minutes)

Minute Paper Activity <u>953677106101</u> * Instruction to the assembler. * convert assembly language to machine language * attach value to variable. * allocate storage location * It is also called pseudo instruction or pseudo opcode	MINUTE PAPER ACTIVITY ASSEMBLER DIRECTIVES :- 1) Converts Assembly language into Machine language 2) PROCEDURE is also called subprogram which involves, 3) CALL and RET function.
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Animation Video for Interrupts and Interrupt Service Routines on 11.01.2020 (10 Minutes)



Significance of Results: (Assessment of Effectiveness/Success of the Activity):

- **Zero Minute Speech:** With interest, most of the students delivered the topics with examples and answered the questions from classmates.
- **Minute Paper:** Helps the students to recall the important assembler directives and their functions.
- **Animation:** Response of 8086 microprocessor for the different interrupts are easily understood by the students through a step by step process.

Changes required for future (if required):

- Students requested to give topics before a day for their preparation for zero minute speech.
- Sometime need to explain a few concepts during animation video presentation with real-time example so more time has to allocate for this activity in the future.

Reflective Critique:

Challenges:

- **Zero Minute Speech:** During zero minute speech few students struggled to deliver the topic and few students hesitated to participate in the activity.
- **Minute Paper:** Not able to involve all the students because few students struggled to recall the points.
- **Animation:** Need to give real-time examples for their better understanding of the topic.

Benefits:

- **Zero Minute Speech:**
 - Able to recall few topics quickly and students can able to link the topic with real-time examples.
- **Animation:**
 - The animation is eye-catching, It is useful when quickly getting and holding the audience's attention.
 - The animation can show the imagined objects in the gesticulation, It is ideal for demonstrating processes.
 - Interactive learning with live-action animation keeps learners interested and reinforces skills.
- **Minute Paper:**
 - Provide a “conceptual bridge” between successive class periods.
 - Improve the quality of class discussion by having students write briefly about a concept or issue before they begin discussing it.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	1	1	1	-	2	-	1	-	1

CO1: The students will be able to describe the architecture of microprocessor 8086 and execute programs based on 8086 microprocessor.

References:

- ❖ <http://www.icaltefl.com/just-a-minute-speaking-activity>
- ❖ <https://busyteacher.org/20564-5-steps-to-making-fluency-fun-just-a-minute.html>
- ❖ <https://oncourseworkshop.com/self-awareness/one-minute-paper/>
- ❖ <https://www.edutopia.org/blog/design-thinking-one-minute-papers-ashley-nahornick>
- ❖ <https://www.youtube.com/watch?v=NXI7NTE95QY>
- ❖ Yu-Cheng Liu, Glenn A. Gibson, “Microcomputer Systems: The 8086 / 8088 Family - Architecture, Programming and Design”, Second Edition, Prentice Hall of India, 2007.